

Membranous Glomerulonephritis

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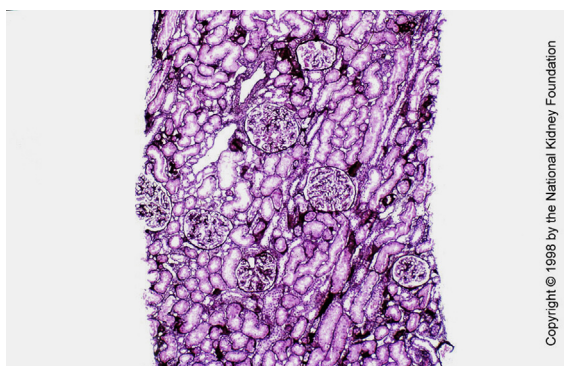


Fig 1. Intact cortex with apparently normal glomeruli on low power in early membranous glomerulonephritis (Jones' silver stain, original magnification $\times 100$).

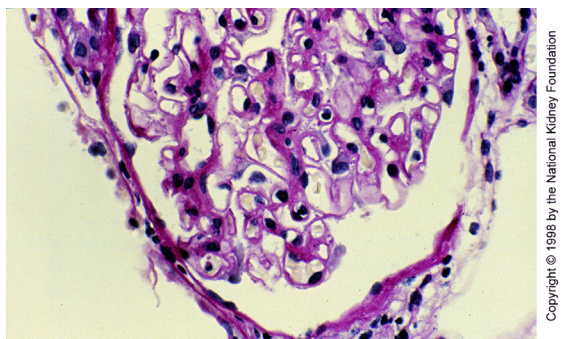


Fig 2. In this case of stage I membranous glomerulonephritis, the capillary wall is slightly prominent and appears more rigid than normal. However, deposits cannot be directly visualized on this periodic acid-Schiff stain (original magnification $\times 400$).

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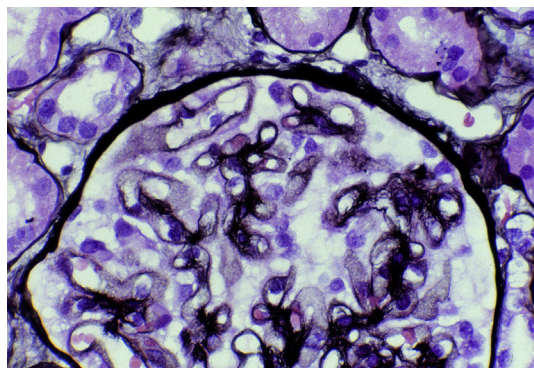
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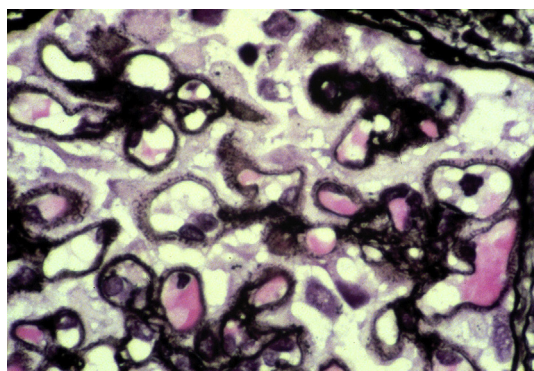
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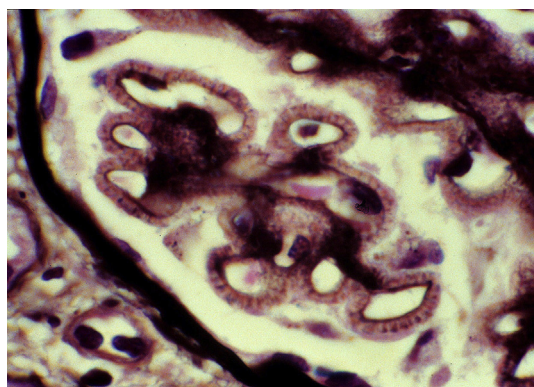
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Fig 3. The thickened capillary wall shows numerous "holes" in tangential sections, indicating deposits. (Deposits do not take up the silver stain.) Well-developed spikes around the deposits are not present in this early stage II membranous glomerulonephritis, but segmental, small nubs of silver-stained basement membrane material protruding from the basement membrane contour on the epithelial side can be seen (Jones' silver stain, original magnification $\times 400$).



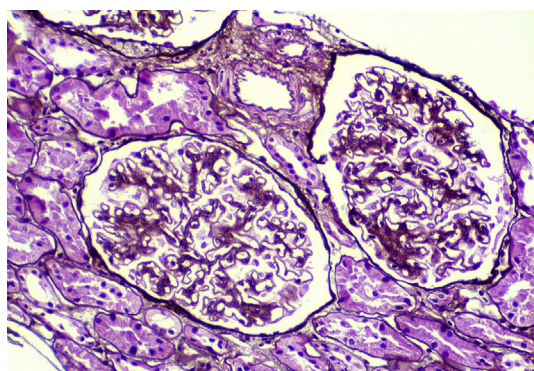
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Fig 4A. High-power oil-immersion view of the markedly thickened capillary wall in (A) stage II-III and (B) stage III membranous glomerulonephritis. Well-developed spikes of basement membrane silver-staining material protrude from the basement membrane. In tangential sections, holes are seen, indicating the presence of the silver-negative deposits. In stage III, some of the deposits are completely surrounded by basement membrane, and the basement membrane appears split in these areas (Jones' silver stain, original magnification $\times 1,000$).



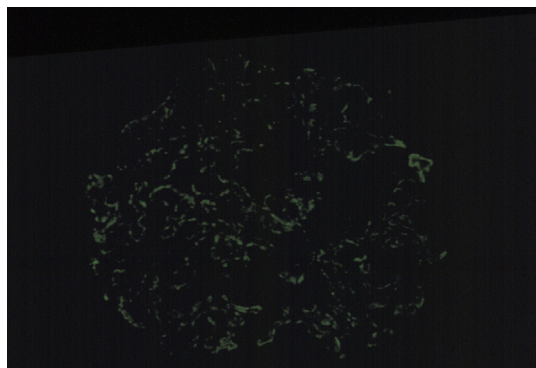
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Fig 4B. High-power oil-immersion view of the markedly thickened capillary wall in (A) stage II-III and (B) stage III membranous glomerulonephritis. Well-developed spikes of basement membrane silver-staining material protrude from the basement membrane. In tangential sections, holes are seen, indicating the presence of the silver-negative deposits. In stage III, some of the deposits are completely surrounded by basement membrane, and the basement membrane appears split in these areas (Jones' silver stain, original magnification $\times 1,000$).



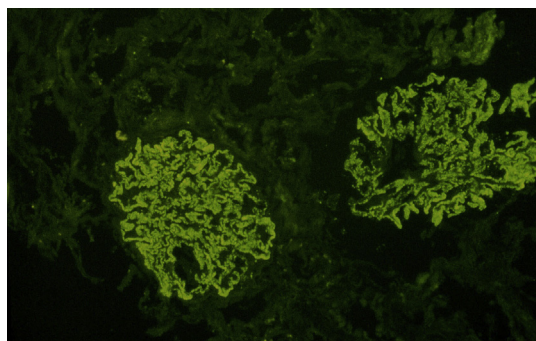
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Fig 5. Idiopathic membranous glomerulonephritis is a diagnosis of exclusion. In this case of secondary membranous glomerulonephritis, hepatitis B infection was the likely underlying etiology. Morphological clues in this case indicating that the membranous glomerulonephritis is secondary rather than idiopathic, include mesangial expansion and mesangial deposits (Jones' silver stain, original magnification $\times 200$).



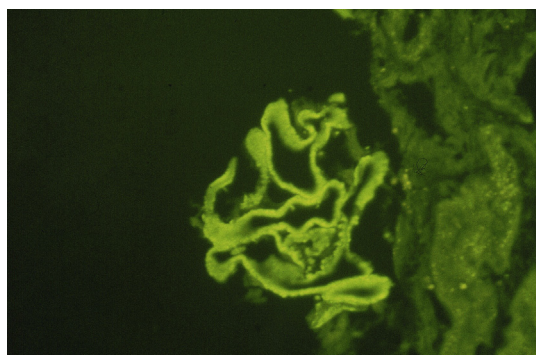
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Fig 6. Diffuse finely granular capillary wall IgG deposits in stage I membranous glomerulonephritis (immunofluorescence with anti-IgG, original magnification $\times 200$).



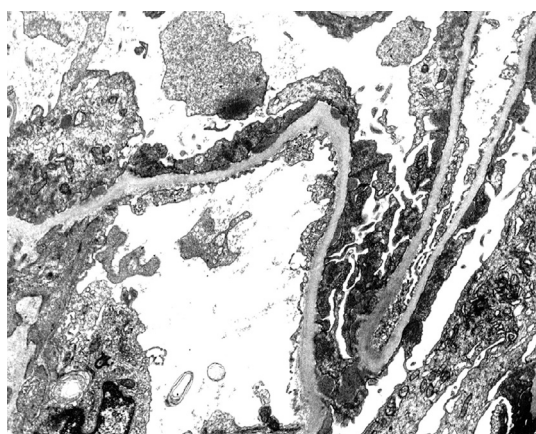
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Fig 7. Diffuse coarsely granular capillary wall IgG deposits in stage II-III membranous glomerulonephritis (immunofluorescence with anti-IgG, original magnification $\times 100$).



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Fig 8. Capillary loops with diffuse, granular capillary wall IgG deposits in stage II-III membranous glomerulonephritis (immunofluorescence with anti-IgG, original magnification $\times 400$).



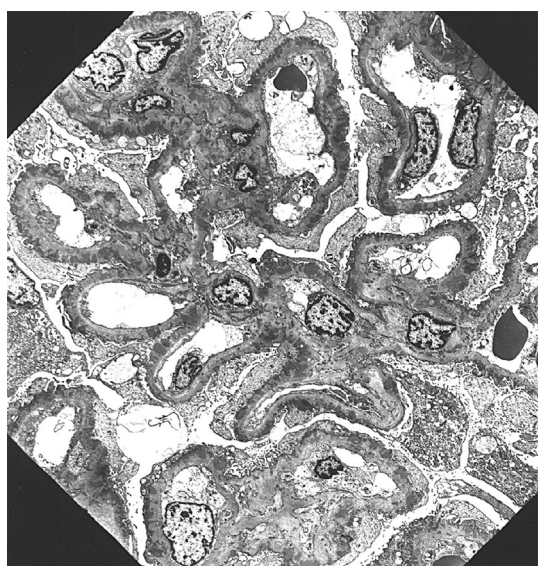
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Fig 9. Small subepithelial dense deposits in early stage I membranous glomerulonephritis. There is no surrounding basement membrane reaction. Foot processes are extensively effaced (transmission electron microscopy, original magnification $\times 10,500$).



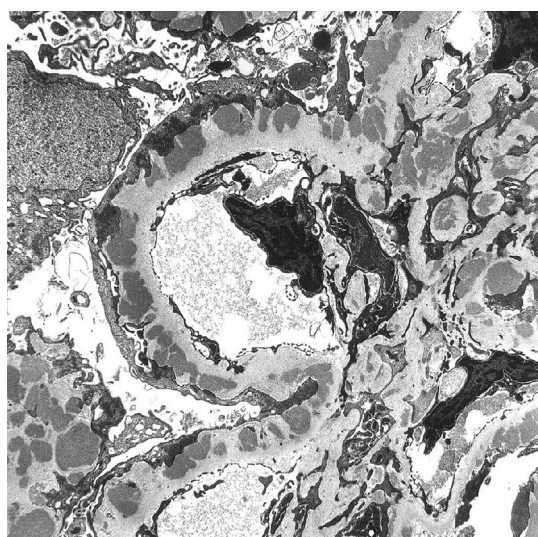
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Fig 10. Medium-sized dense deposits on the epithelial side of the basement membrane (subepithelial) with occasional early basement membrane reaction, late stage I membranous glomerulonephritis (transmission electron microscopy, original magnification $\times 10,200$).



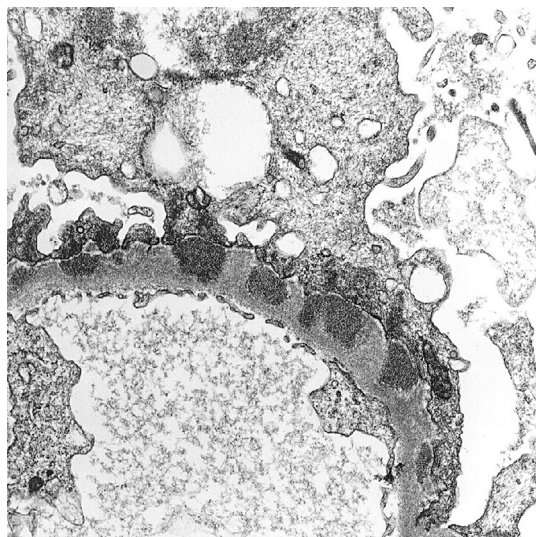
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Fig 11. Stage II membranous glomerulonephritis with many subepithelial electron dense deposits along the capillary loops. Foot processes are effaced. There is basement membrane-like material around the sides of the deposits, which gives rise to the "spike" appearance on Jones' silver stain (see Fig 4) (transmission electron microscopy, original magnification $\times 1,650$).



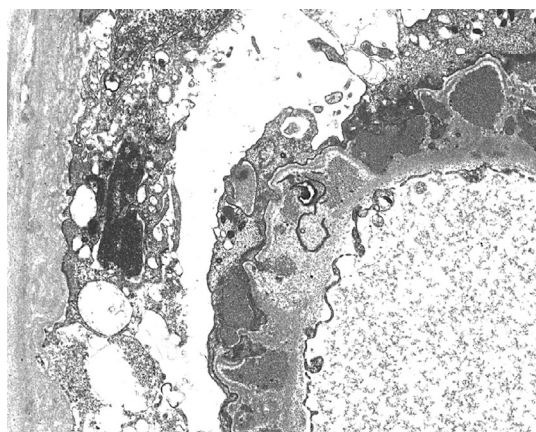
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Fig 12. Stage II-III membranous glomerulonephritis with thickened capillary wall resulting from numerous medium to large subepithelial dense deposits and basement membrane reaction. Basement membrane-like material surrounds some of the deposits, a feature of stage III membranous glomerulonephritis, whereas most of the deposits still only have basement membrane reaction around the side (stage II lesion) (transmission electron microscopy, original magnification $\times 1,650$).



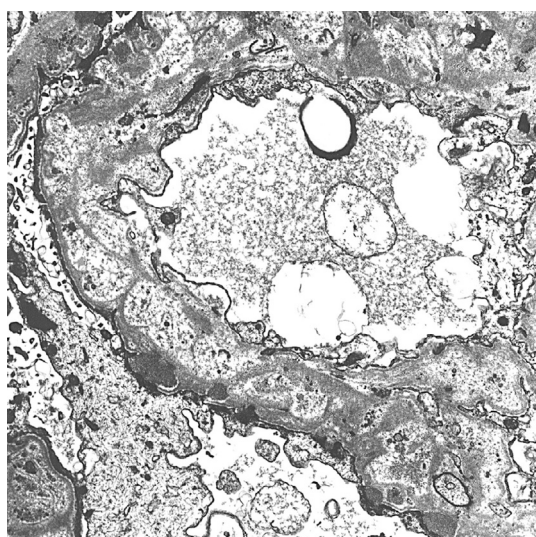
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Fig 13. Stage II membranous glomerulonephritis with medium-sized subepithelial dense deposits and basement membrane reaction around the sides of the deposits, which gives rise to the "spike" appearance on Jones' silver stain (see Fig 4). Epithelial cells show effaced foot processes. The endothelial cells show normal fenestrae (transmission electron microscopy, original magnification $\times 1,650$).



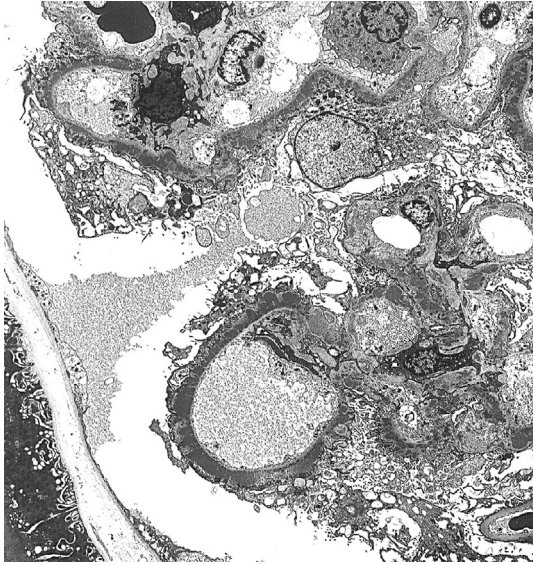
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Fig 14. Stage III membranous glomerulonephritis with medium-sized subepithelial dense deposits and basement membrane reaction surrounding most of the deposits. In the center, a lucent area partially replaces a deposit that is being resorbed. Bowman's capsule and the parietal epithelium is on the far right. (transmission electron microscopy, original magnification $\times 11,000$).



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Fig 15. Stage III-IV membranous glomerulonephritis with mostly rarefied deposits or lucent areas instead of dense deposits, with surrounding and overlying basement membrane reaction. A few small, dense deposits immediately under the epithelium overlying the lucent areas surrounded by basement membrane reaction are present, indicating ongoing, new deposit formation (transmission electron microscopy, original magnification $\times 7,200$).



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Fig 16. Secondary membranous glomerulonephritis with numerous subepithelial deposits (stage II) and mesangial deposits. In this case, the likely underlying etiology was hepatitis B infection (see Fig 5 above) (transmission electron microscopy, original magnification $\times 1,650$).